

Prem Babu Kanaparthi

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Summary

AI/ML and Software Engineer building production LLM, full-stack, retrieval, evaluation, and classical ML systems with measured latency, cost, reliability, and quality gains.

Skills

- AI and Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Prompt Engineering, PyTorch, Scikit-learn, XGBoost
- Evaluation and Experimentation: LLM Evaluation, Eval Harnesses, Retrieval Evaluation, Drift Monitoring, Responsible AI
- Programming and APIs: Python, JavaScript, SQL, FastAPI, Node.js, Express.js, REST APIs, PostgreSQL
- Deployment and Retrieval: LiteLLM, AWS Bedrock, SageMaker, Docker, CloudWatch, FAISS, BM25, MCP

Experience

Rochester Institute of Technology Jan 2026 - Apr 2026

Graduate Researcher . Rochester, NY

- Extended Stanford's Generative Agents with a 72-feature LSTM predictive model, controlled scenarios, ablations, and statistical comparisons across 205,940 agent-step records.
- Validated statistically significant emergent emotion patterns against Gallup 2024 human baselines and documented model behavior, limitations, and failure modes (poster).

Concentrix + Webhelp

Feb 2024 - Jul 2024

Generative AI Engineer . Newark, CA

- Owned the production LLM inference routing layer for 50K daily requests across 3 foundation models on AWS Bedrock and SageMaker; used LiteLLM, provider fallback, and latency-aware routing to cut p95 latency from 4s to 1.5s.
- Built production LLM evaluation and safety gates with CloudWatch logging, hallucination checks, drift monitoring, and a 500-case test set; reduced incidents by 42% and MTTD by 35%.
- Reduced monthly inference spend from \$45K to \$37K through cost-aware routing, prompt caching, and provider fallback while maintaining 95%+ task success.

AlphaBits Technologies

Aug 2023 - Jan 2024

Data Science Intern . Bengaluru, India

- Rebuilt a Python search-ranking experimentation pipeline with preprocessing, feature generation, SQL-backed analysis, and 5 model variants; cut iteration time by 90% and improved relevance by 10%.
- Created an offline ranking evaluation harness tracking relevance, error cases, and preprocessing differences across model variants before deployment.

iNeuron AI

Jun 2023 - Aug 2023

ML Engineer Intern . Bengaluru, India

- Developed XGBoost classifiers for learner dropout risk (0.86 AUC across 12,000 learners) and phishing URL detection (20+ URL/domain features, 92% accuracy).
- Built a behavioral feature pipeline converting raw activity logs into 24 rolling-window features with drift and quality checks for retraining.

Exposys Data Labs

May 2021 - Jun 2022

Software Developer Intern . Bengaluru, India

- Built reusable React/Bootstrap interfaces and Node.js/Express REST APIs for an internal inventory system backed by PostgreSQL; client-side validation reduced order-entry errors by approximately 25%.
- Automated weekly reporting with SQL and assisted API refactoring and database indexing that improved retrieval speed by approximately 20%; added Jest/Mocha tests with over 70% coverage on assigned modules.

Education

Rochester Institute of Technology

Aug 2024 - May 2026

Master of Science in Artificial Intelligence, completed May 2026 . Rochester, NY

National Institute of Technology Silchar

Aug 2020 - May 2024

Bachelor of Technology in Computer Science, completed May 2024 . Silchar, India

Projects

QuanTRAG - Grounded Financial Research Copilot

GitHub

- Built cited retrieval over SEC filings with 4 retrieval modes (dense, sparse, hybrid, and reranked), hallucination checks, abstention, metadata filtering, and a 52-question benchmark.

Nori - AI Job Discovery and Automation Platform

norinote.xyz

- Built a 0-to-1 platform with bounded async scrapers, ATS adapters, Redis queues, sentence-transformer ranking, and scheduled workflows backed by a 31,000+ company ATS registry; surfaced 120+ roles/day.